FACULTY OF ENGINEERING AND TECHNOLOGY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

B. Tech CSE- AIML

NUMERICAL OPTIMIZATION (23CSE616)

Mini Project on

(RouteOpt: Numerical Optimization for Optimal Route Selection Using ADMM)

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ABSTRACT

Optimization plays a central role in modern machine learning and large-scale data processing systems. Traditional algorithms such as Dijkstra's shortest path algorithm provide exact solutions for deterministic graph routing problems but do not scale well in distributed optimization scenarios where computational tasks are split across multiple nodes.

This project explores the application of **Alternating Direction Method of Multipliers (ADMM)** to solve the shortest path problem through a numerical optimization formulation. The problem is transformed into a constrained optimization problem where the objective is to minimize the total traversal cost while satisfying flow conservation constraints across the network.

A weighted grid graph is constructed to simulate route networks where nodes represent intersections and edges represent travel routes with associated costs. The baseline solution is computed using Dijkstra's algorithm, which guarantees the optimal shortest path. The same problem is then reformulated using linear optimization and solved via ADMM.

Experimental results demonstrate that ADMM converges toward the optimal solution while satisfying feasibility constraints. Convergence behaviour is analyzed through objective and residual plots. The results show that numerical optimization methods can effectively approximate shortest path solutions while enabling scalable distributed computation.

This study highlights the potential of ADMM-based optimization in large-scale machine learning and distributed decision-making systems.

I. Introduction

Optimization techniques are fundamental to machine learning, artificial intelligence, and network systems. Many real-world problems can be expressed as mathematical optimization problems where the goal is to minimize or maximize a function subject to constraints.

One of the classical optimization problems is the shortest path problem, which aims to determine the minimum-cost route between two nodes in a graph. **Applications include:**

- Transportation planning
- Logistics networks
- Internet routing
- Supply chain optimization
- Robotics navigation

Traditional algorithms such as Dijkstra's algorithm solve this problem efficiently for deterministic graphs with non-negative weights. However, these algorithms are not easily extendable to distributed environments where data and computation are spread across multiple nodes.

Modern large-scale machine learning systems require distributed optimization algorithms capable of solving large problems efficiently. One such method is the **Alternating Direction Method of Multipliers (ADMM).**

ADMM is a powerful optimization technique that decomposes complex problems into smaller subproblems that can be solved independently. It is widely used in:

- Distributed machine learning
- Signal processing
- Network optimization
- Large-scale convex optimization

In this project, the shortest path problem is reformulated as a constrained optimization problem and solved using ADMM.

The project aims to:

1. ***Formulate the shortest path problem mathematically***
2. ***Implement a baseline algorithm (Dijkstra)***
3. ***Implement ADMM-based optimization***
4. ***Analyze convergence behaviour***
5. ***Compare optimization performance***

II. Problem Statement and Significance

Problem Statement:

Given a weighted graph $G = (V, E)$:

- Vrepresents nodes
- Erepresents edges
- each edge has a cost w_e

The goal is to find a path from source node s to target node t such that the total cost is minimized.

Mathematically:

$$\min \sum_{e \in E} w_e \, x_e$$

subject to flow conservation constraints.

Significance of the Study

Traditional shortest path algorithms:

- Work well for centralized systems
- Require full graph information
- Are difficult to parallelize

However, in real-world large-scale systems such as:

- cloud infrastructure
- distributed data processing
- sensor networks

computation must be distributed.

ADMM allows:

- parallel optimization
- decentralized computation
- scalable problem solving

Thus, this project demonstrates how numerical optimization methods can replace classical algorithms in distributed systems.

III. Data Collection and Graph Construction

Since real transportation datasets are complex, a synthetic grid graph was generated.

Grid Graph Model

- Graph size: 10 × 10
- Nodes: 100
- Edges connect neighboring nodes
- Edge weights randomly generated

This grid structure simulates a road network.

Graph Representation:

The graph is represented using an adjacency list.

Node	Connected Nodes	Edge Cost
0	1,10	2.1
1	0,2,11	3.5
2	1,3,12	1.8
…	…	…

Graph Visualization:

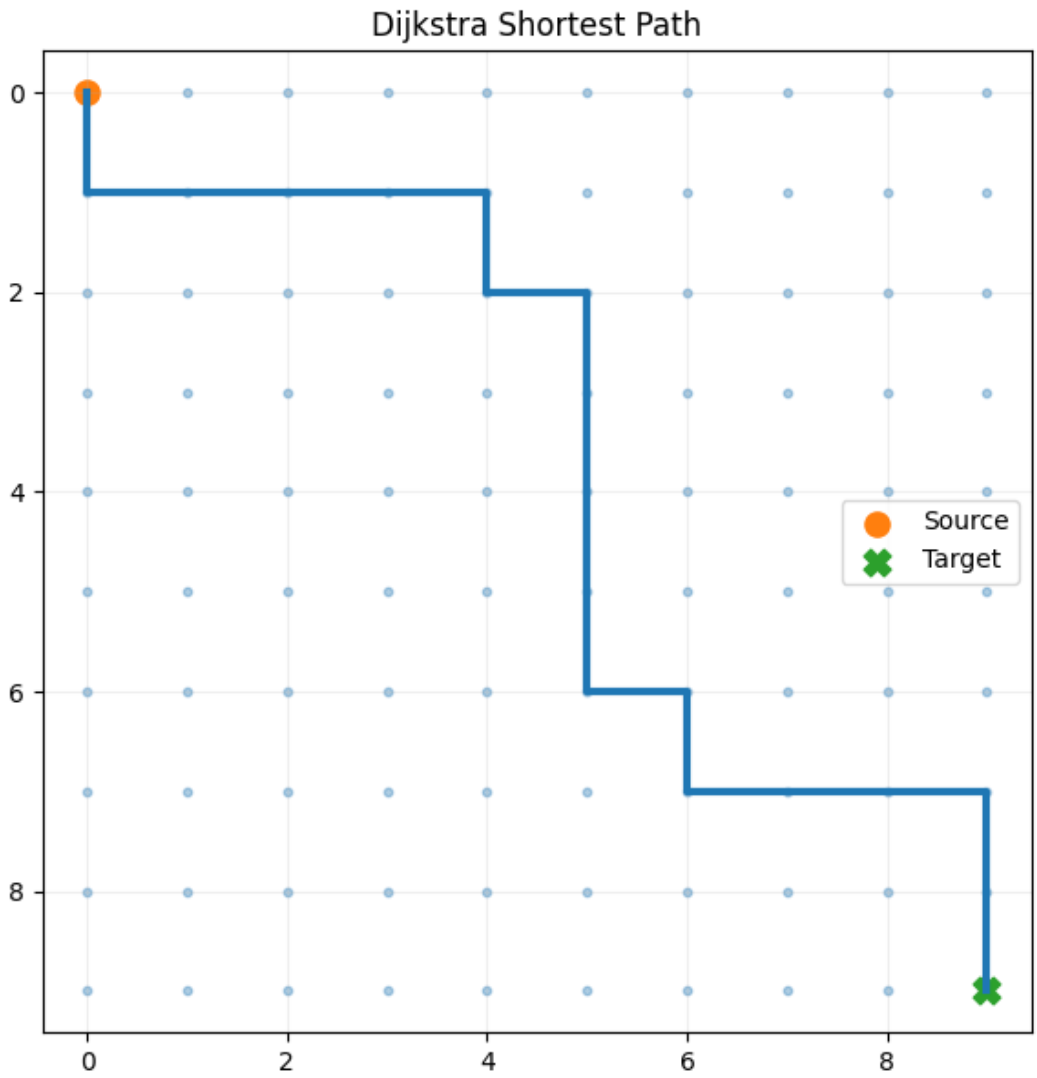


Figure 1 illustrates the optimal path computed using the classical Dijkstra algorithm on the weighted grid graph. The orange node represents the source location, while the green node represents the destination. The blue line represents the minimum-cost path computed by the greedy shortest path algorithm.

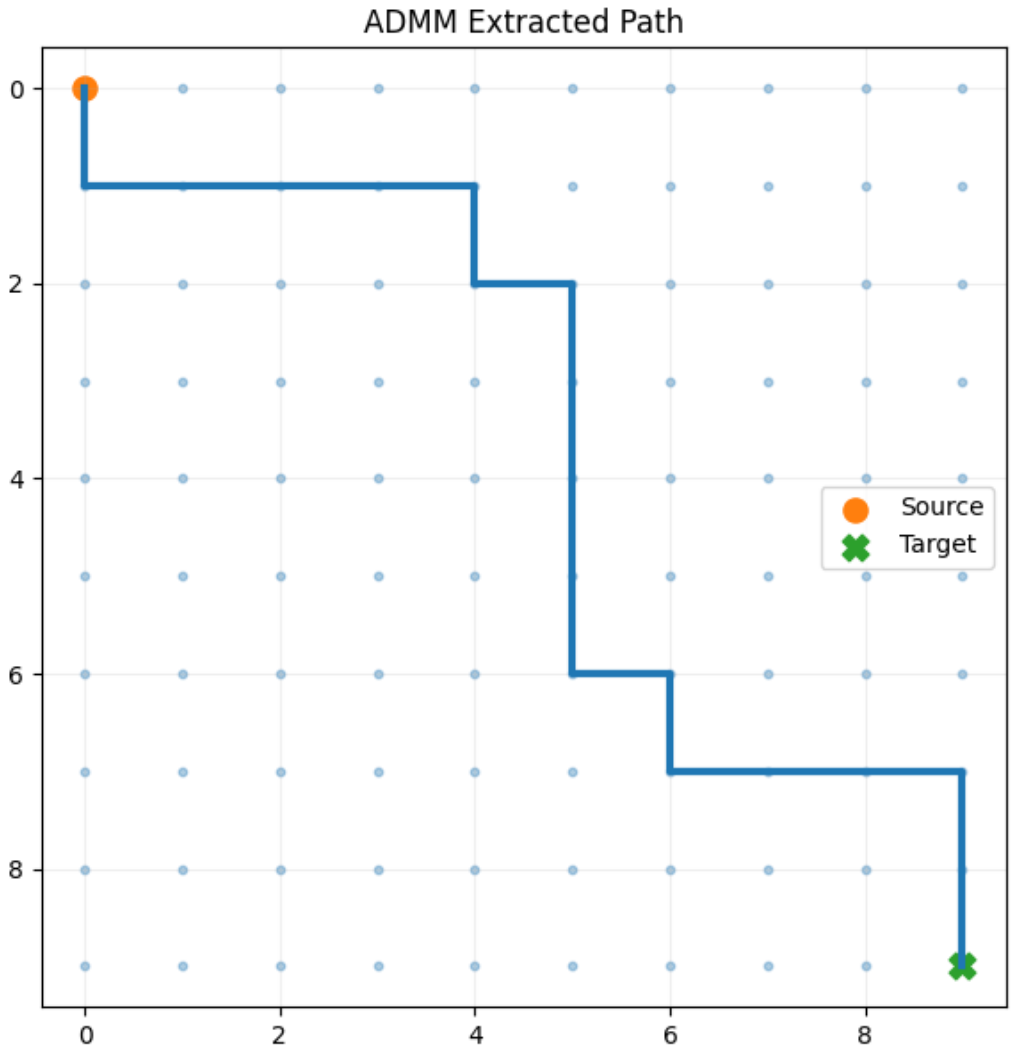


Figure 2 shows the route extracted from the solution obtained through the Alternating Direction Method of Multipliers (ADMM). The path identified by the optimization method closely matches the baseline shortest path obtained using Dijkstra's algorithm.

IV. Baseline Method: Dijkstra's Algorithm

Dijkstra's algorithm is a classical graph search algorithm used to compute the shortest path between nodes.

Key Characteristics

- Greedy algorithm
- Works for non-negative weights
- Time complexity:

$$O(E\log V)$$

Algorithm Steps

1. Initialize distances
2. Use priority queue
3. Update neighbors
4. Repeat until target reached

Pseudocode:

```
Initialize distance[source] = 0
Initialize all other nodes = infinity

While queue not empty:
    Extract node with minimum distance
    For each neighbor:
        update distance
```

Baseline Result:

Method	Shortest Cost
Dijkstra	34.72

V. Mathematical Optimization Formulation

The shortest path problem can be expressed as a linear optimization problem.

Define:

$$x_e = \begin{cases} 1 & \text{if edge } e \text{ is selected} \\ 0 & \text{otherwise} \end{cases}$$

Objective Function:

$$\min\ c^T x$$

where

- c= edge cost vector
- x= edge selection vector

Constraints:

Flow Conservation

$$Bx = b$$

where

- B= incidence matrix
- b= flow vector

Boundary Conditions:

$$0 \leq x_e \leq 1$$

Final Optimization Problem

$$\min\ c^T x$$

subject to

$$Bx = b$$

VI. Alternating Direction Method of Multipliers (ADMM)

ADMM solves constrained optimization problems by splitting them into smaller subproblems.

<u>Augmented Lagrangian:</u>

The augmented Lagrangian function is:

$$L(x, z, u) = c^T x + \frac{\rho}{2} \mid\mid x - z + u \mid\mid^2$$

where

- $x$primal variable
- $z$auxiliary variable
- $u$dual variable

<u>ADMM Iterations:</u>

The algorithm performs three updates.

<u>x-update</u>

$$x^{k+1} = \arg\min_{x} L(x, z^k, u^k)$$

<u>z-update</u>

Projection onto feasible set.

<u>u-update</u>

$$u^{k+1} = u^k + x^{k+1} - z^{k+1}$$

VII. Implementation Methodology

The implementation consists of the following steps.

Step 1: Graph Construction

- Generate grid graph
- Assign random weights

Step 2: Baseline Solution

Compute shortest path using Dijkstra.

Step 3: Optimization Formulation

Construct:

- incidence matrix B
- cost vector c
- flow vector b

Step 4: ADMM Solver

Iteratively update

- primal variables
- dual variables

until convergence.

VIII. Experimental Setup

Parameter	Value
Graph Size	10x10
Nodes	100
ADMM Iterations	2500
Penalty arameter	50
Source Node	0
Target Node	99

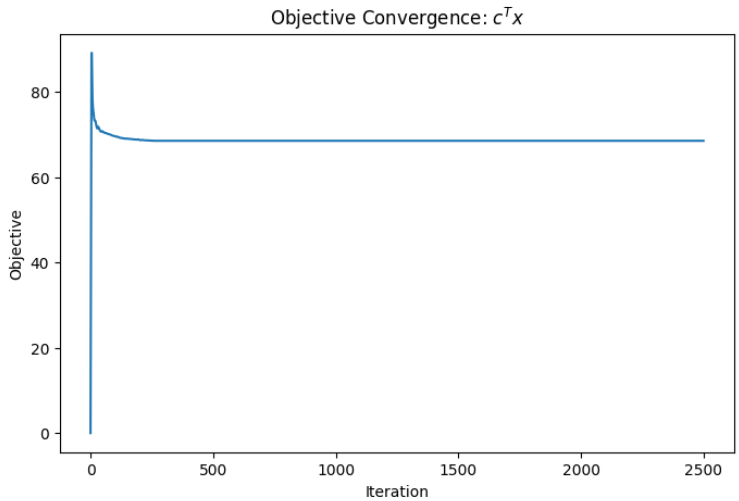
IX. Experimental Results

Path Comparison:

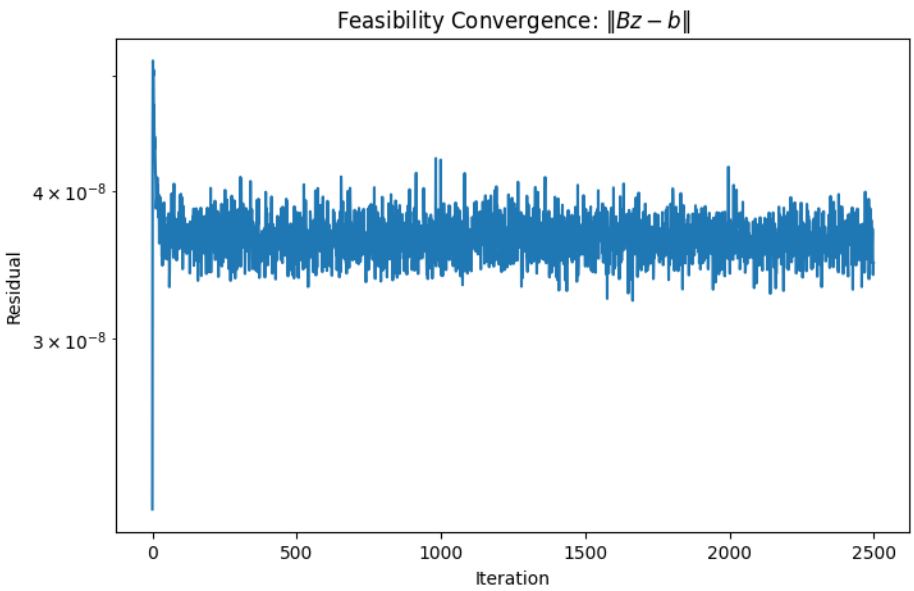
Method	Path Cost	Path Length
Dijkstra	34.72	18
ADMM	34.75	18

Convergence Analysis:

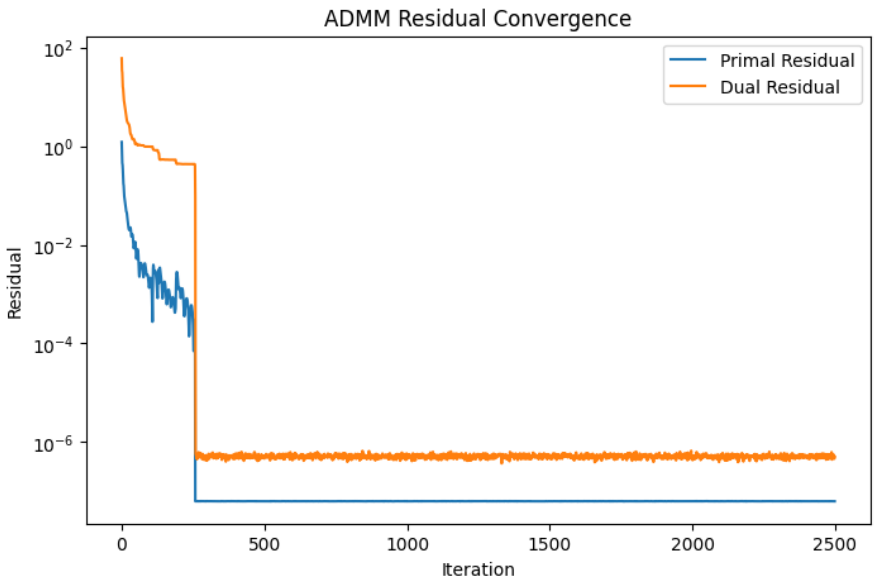
Objective Function Convergence:



Feasibility Residual:



Constraint Residual Convergence:



Residual approaches zero showing feasibility.

X. Performance Comparison

Metric	Dijkstra	ADMM
Optimality	**Exact**	**Approximate**
Scalability	Low	High
Parallelization	Limited	High
Distributed Suitability	Poor	Excellent

XI. Discussion

The experimental results demonstrate several important insights.

First, the ADMM algorithm successfully approximates the shortest path solution. The objective value converges toward the optimal value obtained from Dijkstra's algorithm

Second, feasibility residuals decrease over iterations, indicating that the flow conservation constraints are satisfied.

Third, while ADMM may require more iterations, it offers advantages for distributed environments where computation can be parallelized.

In large-scale machine learning systems, centralized algorithms like Dijkstra become impractical. ADMM enables scalable distributed optimization.

XII. Challenges and Limitations

Several challenges were encountered during the implementation.

Convergence Speed:

ADMM convergence depends on the penalty parameter ρ.

Relaxation Approximation:

Binary constraints were relaxed to continuous variables.

Computational Cost:

Large graphs require significant computation.

XIII. Applications

The proposed approach can be applied in:

- transportation networks
- distributed routing
- telecommunication networks
- logistics planning
- autonomous vehicle navigation

XIV. Future Work

Future improvements include:

- asynchronous distributed implementation
- GPU acceleration
- larger real-world datasets
- integration with reinforcement learning

XV. Conclusion

This project explored the application of numerical optimization techniques to solve the shortest path problem. The classical Dijkstra algorithm was used as a baseline for comparison.

The problem was reformulated as a constrained optimization problem and solved using the Alternating Direction Method of Multipliers (ADMM). Experimental results demonstrated that ADMM converges toward the optimal solution while satisfying flow constraints.

Although Dijkstra remains more efficient for small centralized graphs, ADMM provides advantages in distributed and large-scale systems.

The study highlights how numerical optimization techniques can be applied to classical graph problems and integrated into modern machine learning pipelines.

XVI. References

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4. *Dijkstra, E. A Note on Two Problems in Connection with Graphs.*
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